Linear algebra: Exercise Chapter 7

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1. Compute eigenvalues and eigenvectors of the following matrices.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 3 & -19 & 11 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

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2. Let $2, -3, 2, 5$ are the eigenvalues of 4×4 matrix A. Find the determinant and trace of A.

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3. Let λ is the eigenvalue of $n \times n$ matrix A corresponding the eigenvector x. Prove the following statements:
   (a) λ^2 is an eigenvalue of A^2 corresponding the eigenvector x.
   (b) If A is invertible, then λ^{-1} is an eigenvalue of A^{-1} corresponding the eigenvector x.
   (c) For each $\mu \in \mathbb{R}$, $\lambda + \mu$ is an eigenvalue of $A + \mu I$ corresponding the eigenvector x.
   (d) For each $\mu \in \mathbb{R}$, $\mu\lambda$ is an eigenvalue of μA corresponding the eigenvector x.

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4. Consider the matrix A in question 1. Compute the eigenvalues and eigenvectors of matrices $A - 3I$, $-2A$, A^{-1}, A^2.

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5. Compute $A^5 x$ for the following matrix A and vector x:

$$A = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 10 & -4 \\ 0.5 & 12 & -4 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 4 \\ 7 \end{bmatrix}.$$